I chose a 3D scene project I did for my CS330 course. This project was to make a 3D scene and I made one with a table, plate, coffee cup, and coffee and added texture to everything and some lighting. I chose this because it seemed like a good example of what I had to do in the course and there were a lot of parts in the code that I could improve upon.

I focused mainly on making the structure of the code better by reducing a lot of the repeated code. In the original there was a lot of blocks of code that were being repeated to set up different objects and lights in the scene so instead I just made reusable functions so that the set up didn't have to be written so many times. This made the code for rendering and lighting shorter and a lot more organized while still keeping the scene the same. I also improved error handling for loading textures and finding materials so that problems can be found easier if something doesn't load correctly.

This artifact shows my skills because I was able to review an older project and find ways it can be improved instead of just adding features. I focused on making the code easier to read and more organized. I think the enhancements helped meet the course outcome because I was able to show my skills in software engineering and design by improving the structure and organization. I did not make any changes to the outcome I planned originally because the enhancements I did were still based on the areas I found during my code review.

A challenge I did have was testing the improved project. I did the enhancements as best as I could with what I had but it's unable to be tested because when I ended up trying to run it I realized the version of the project I had saved on GitHub did not include all of the dependencies from the original project so some of the supporting files and libraries needed to build the program were missing so I wasn't able to do a full test on the changes I made. I was still able to review the source code and complete the improvements I planned though.

Working on this helped me understand how much of a difference code structure can actually make and even though the original project did what was required, there were still ways to make the code cleaner and reduce all unnecessary code. I also learned that its important to make sure all files and dependencies needed to run the project are saved with your project in case you need to go back to it later.